A LIGHTWEIGHT STATIC ANALYZER FOR VALIDATING LATEX-BASED DISSERTATIONS AGAINST UNIVERSITY FORMATTING SPECIFICATIONS

By

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A DISSERTATION PRESENTED TO THE GRADUATE SCHOOL
OF THE UNIVERSITY OF FLORIDA IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF
DOCTOR OF PHILOSOPHY

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To every graduate student who discovered a formatting rule the night before submission — you are almost there.

ACKNOWLEDGEMENTS

If you are reading this in your own dissertation, take a breath. You have done the hard part. The remaining work is mechanical: get the formatting right, hit submit, then sleep.

I thank my advisor for many patient conversations about scope, my committee for their feedback, and the UF Graduate Editorial Office whose published formatting standards made this work both necessary and possible. I am also grateful to the Thesis & Dissertation Support Center at UF Information Technology, whose template authors deserve credit for encoding most of the formatting rules directly into the LaTeX class so students can spend their time on the research and not on margins.

To the future reader who has copied this acknowledgements file as a starting point: replace this prose with your own and remember to thank the people who actually got you here. They earned it. So did you.

This work was supported in part by departmental teaching and research assistantships.

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LIST OF FIGURES

LIST OF OBJECTS

LIST OF ABBREVIATIONS

ETD Electronic Thesis or Dissertation

GIMS Graduate Information Management System

LoA List of Abbreviations (this list)

LoF List of Figures

LoO List of Objects

LoT List of Tables

PDF Portable Document Format

PDF/UA Portable Document Format / Universal Accessibility

ToC Table of Contents

UF University of Florida

WCAG Web Content Accessibility Guidelines

Abstract of Dissertation Presented to the Graduate School
of the University of Florida in Partial Fulfillment of the
Requirements for the Degree of Doctor of Philosophy

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May 2026

Chair: Chair Example
Cochair: Co-chair Example
Major: Computer Science

Graduate schools enforce strict formatting rules on doctoral dissertations, but the rules are scattered across web pages, PDF guides, and template documentation. Students who already compile their dissertation locally still face a tedious manual checklist before submission, often performed under time pressure near the term deadline. A formatting mistake caught after the first submission deadline can delay graduation by a full semester.

This dissertation presents a lightweight static analysis tool that validates LaTeX-based dissertations against the University of Florida Graduate Editorial Office's formatting specification. The tool operates in two complementary layers. A source layer inspects the LaTeX project for overrides of the official UF template's defaults — for example, a manually inserted `\pagenumbering{roman}` that would violate the new template's arabic-throughout rule. A PDF layer inspects the rendered output for properties the source cannot predetermine: actual margins per page, embedded font families, page numbering scheme as rendered, and tagged-structure presence for accessibility compliance. The two layers are designed to complement each other: source-layer findings carry precise location data; PDF-layer findings carry rendered-truth confidence.

The analyzer accepts project archives, project directories, or compiled PDFs as input. Output is a grouped, severity-tiered report with each finding citing the originating UF rule. A

demo dissertation that satisfies every rule serves as both regression fixture and pedagogical reference for new graduate students learning the template.

This work focuses on doctoral dissertations using the Fall 2025+ UF template. Master's theses and the older template (accepted through Summer 2026) are out of scope for the initial release. The source-layer checks are expected to recover the majority of formatting violations recorded in the Editorial Office's published rejection patterns, with the PDF layer adding defense-in-depth against template bypass scenarios. A systematic evaluation on a corpus of public UF dissertations is planned for a follow-up study.

CHAPTER 1 INTRODUCTION

Doctoral students at the University of Florida must submit their dissertations through the Graduate Information Management System (GIMS). Before submission, the dissertation must conform to a detailed formatting specification maintained by the Graduate Editorial Office. Common reasons for rejection include incorrect margins, omitted required sections, non-functional hyperlinks, and improper use of spacing or fonts.

This dissertation introduces a tool that helps students catch such issues automatically before they trigger a rejection. The work is motivated by three observations: (1) the official UF LaTeX template already enforces most formatting rules by construction, but students can and do override the template; (2) some properties — for example, the rendered margin on a page that includes a wide figure — cannot be determined from source alone; and (3) students near a submission deadline benefit from a fast, offline tool that flags likely problems with references back to the UF rule being violated.

If you are reading this as a soon-to-be doctor: the formatting specification is finite, the rules are mechanical, and the validator is on your side. You wrote the dissertation. The hard part is done.

1.1 Background

The UF Graduate School publishes its formatting specification at `success.grad.ufl.edu/td/formatting/`. The UF IT Help Desk publishes the LaTeX template at `it.ufl.edu/helpdesk/graduate-resources/`. The current template, updated for Fall 2025, requires LuaLaTeX with TeX Live 2025 to enable PDF tagging for accessibility.

1.2 Contributions

This dissertation makes three contributions.

1.2.1 A two-layer validator architecture

A static analyzer that inspects both LaTeX source and rendered PDF. Source-layer checks are precise and localizable to file and line. PDF-layer checks confirm rendered-output truths the source layer cannot predict.

1.2.2 A rule-by-rule catalog with citations

A documented mapping from UF formatting specification entries to machine-checkable detection strategies, published as `docs/uf-rules.md` so that the tool's checks can be audited against the official specification.

1.3 Organization of this Dissertation

Chapter 2 describes the validator's architecture in detail. Chapter 3 summarizes findings and discusses future work. Appendix A lists supplementary rule identifiers, and Appendix B describes the synthetic broken inputs used to verify the validator's coverage.

CHAPTER 2 METHODS

This chapter describes the validator's two-layer architecture. The source layer is presented in Section 2.1 and the PDF layer in Section 2.2. Reporting conventions follow in Section 2.3.

The validator's design follows a standing convention in the UF template documentation [2]:

> We don't recommend that you change much of anything in the class file unless you're absolutely sure of what you are doing.

The validator honors this guidance by treating the template as authoritative and flagging only student-introduced overrides. If you are reading this demo as a reference, take heart: the template does most of the work. Your job is to write a great dissertation and not fight with margins.

2.1 Source-Layer Checks

The source layer parses the LaTeX project files and reports violations that can be determined without compiling the document.

2.1.1 Override Scans

Most UF formatting rules are enforced by the `ufdissertation` class. A source-layer override scan looks for patterns that bypass the class's defaults — for example, a manual `\pagenumbering{roman}` that would violate the new template's arabic-throughout rule per UF-F6 [1].

The override-detection problem reduces to pattern matching, which is formalized in Lemma 2-1.

Lemma 2-1. *Let T be the set of formatting properties enforced by the* `ufdissertation` *class. For each $t \in T$ there exists a finite set O_t of LaTeX command patterns that, if present in a project's source, override the class's enforcement of t. Detection of any violation of t at the source layer reduces to scanning the source for membership in O_t.*

Theorem 2-1. *The source layer is sound: every violation it reports corresponds to a real override of a UF formatting rule. The source layer is not complete: violations introduced via mechanisms outside $\bigcup_t O_t$ are missed.*

The PDF layer (Section 2.2) provides defense-in-depth against the incompleteness identified in Theorem 2-1. Algorithm 2-1 states the procedure formally.

Require: project files F, rule t with override patterns O_t

Ensure: list of findings L

$L \leftarrow [\,]$
for all $f \in F$ with extension `.tex` **do**
    $lines \leftarrow \textsc{ReadLines}(f)$
    for $i \leftarrow 1$ to $|lines|$ **do**
        for all $p \in O_t$ **do**
            if $\textsc{Match}(p, lines[i])$ **then**
                append (f, i, t, p) to L
            end if
        end for
    end for
end for
return L

Object 2-1. Source-layer override scan for a single rule.

2.1.2 Presence Checks

Other rules require the student to set specific macros or files. A presence check verifies, for instance, that `\setAbstractFile` has been called and that the named abstract file exists on disk.

2.2 PDF-Layer Checks

The PDF layer inspects the compiled output for properties the source layer cannot predict.

2.2.1 Geometric Properties

Margins, page numbering placement, and embedded font families are verified against the rendered output. Equation 2-1 formalizes the margin acceptance criterion:

$$m_{\text{page}} \geq 1.0\,\text{in} \quad \forall\, m \in \{m_{\text{top}}, m_{\text{bottom}}, m_{\text{left}}, m_{\text{right}}\}. \tag{2-1}$$

2.2.2 Accessibility Tags

Tagged-structure presence is verified by inspecting the PDF's `/StructTreeRoot`. UF's template, when compiled with LuaLaTeX under TeX Live 2025, produces tagged output by default [2].

2.3 Reporting

Findings are grouped by category and severity. Each finding includes a file and line (source-layer) or page number (PDF-layer), the UF rule identifier, and a URL pointing to the rule's authoritative source. Severity tiers are listed in Table 2-1; per-rule layer assignment in Table 2-2; and a longer enumeration of formatting rules in Table 2-3. The overall architecture is sketched in Figure 2-1, input-mode routing in Figure 2-2, and per-category rule counts in Figure 2-3.

Submitting a dissertation is the last mile of a long race. The validator exists to help you finish that mile without tripping on a shoelace.

Object 2-2. Reference dataset: anonymized UF dissertation submissions, 2018–2025.

Object 2-3. Source archive of the validator implementation accompanying this dissertation.

Table 2-1. Severity tiers used in the validator's report.

Tier	Meaning
must-fix	Documented UF rule violation; submission will be rejected.
review	Likely issue requiring human judgment.

Table 2-3. Documented formatting rules (F-series), severity, and primary detection layer. Continues across page breaks if the table grows.

Rule	Severity	Primary layer
UF-F1 Margins	must-fix	source + pdf
UF-F2 Font family	must-fix	source + pdf
UF-F3 Font size 12pt	must-fix	source
UF-F4 Line spacing	must-fix	source + pdf
UF-F5 Alignment (ragged-right)	must-fix	source
UF-F6 Page numbering	must-fix	source + pdf
UF-F7 Paragraph indentation	must-fix	source

continued on next page

Table 2-3 continued.

Rule	Severity	Primary layer
UF-F8 Required page order	must-fix	source
UF-F9 Singleton structure	must-fix	source
UF-F10 $\geq$ 3 chapters	must-fix	source
UF-F11 Heading style hierarchy	must-fix	source
UF-F12 No mid-chapter blank gaps	review	pdf
UF-F13 `ufdissertation` class	must-fix	source
UF-F14 Required metadata macros	must-fix	source
UF-F15 Abstract $\leq$ 350 words	must-fix	source + pdf
UF-F16 Subsection pairing	review	source

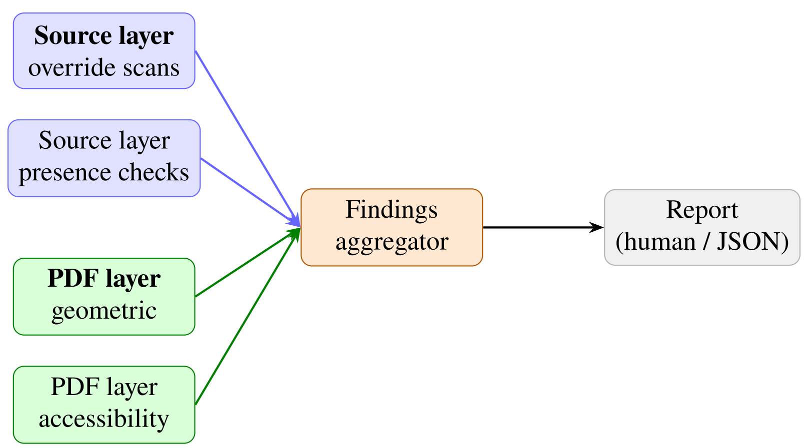

Figure 2-1. High-level architecture of the two-layer validator. The source layer (blue) and the PDF layer (green) feed into a findings aggregator (orange), which produces a grouped report (gray).

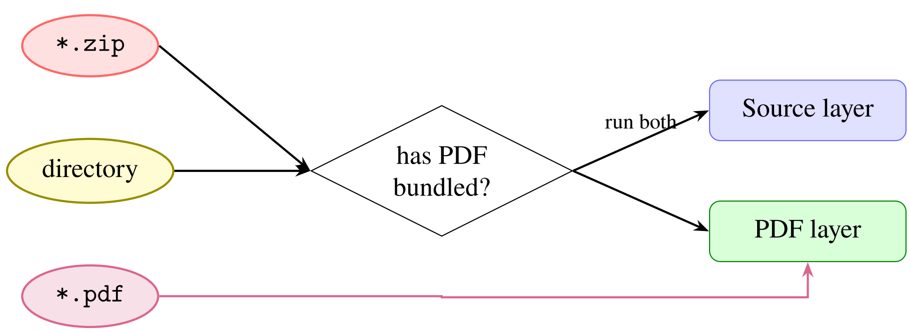

Figure 2-2. Input-mode routing. `.zip` and directory inputs run both layers; if no PDF is bundled, the source is compiled to obtain one. A bare `.pdf` input skips the source layer.

Table 2-2. Per-rule layer assignment (subset).

Rule	Severity	Layer
UF-F1 Margins	must-fix	source[a]+ pdf
UF-F6 Page numbers	must-fix	source[a]+ pdf
UF-F8 Page order	must-fix	source
UF-F12 Blank gaps	review	pdf
UF-F15 Abstract $\leq$ 350 words	must-fix	source + pdf
UF-S3 Internal refs	must-fix	source
UF-A1 PDF tagged	review	pdf

[a] Source layer checks for student-introduced overrides of the template's defaults; the template itself enforces the rule.

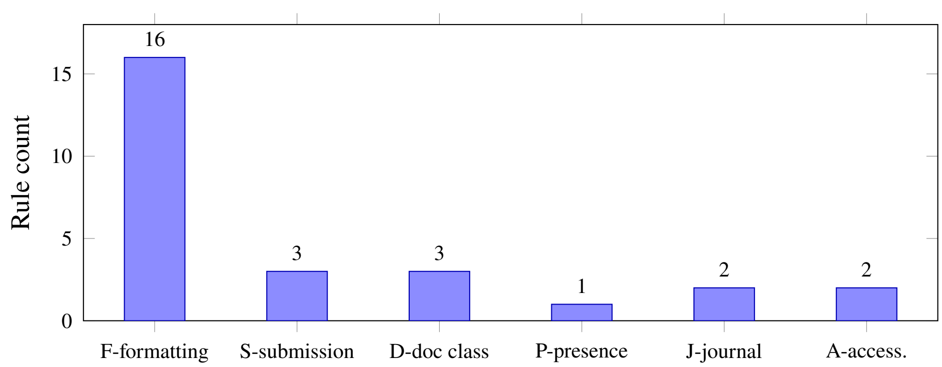

Figure 2-3. Number of documented rules by category. The formatting series dominates the rule catalog; submission, doc-class, presence, journal, and accessibility categories contribute smaller but still meaningful coverage. The submission bar counts three rules rather than the five UF-S identifiers in the catalog: UF-S2 shares a code path with UF-F8 (it is a marketing-friendly grouping in the report output, not a separate check), and UF-S4 (external URL liveness) is deferred to v1.1.

CHAPTER 3 CONCLUSION

This dissertation presented a two-layer validator for the University of Florida doctoral dissertation formatting specification. Chapter 1 motivated the work and outlined contributions. Chapter 2 described the source-layer and PDF-layer checks, formalized the override-detection problem in Lemma 2-1, stated soundness in Theorem 2-1, gave the margin acceptance criterion in Equation 2-1, summarized severity tiers in Table 2-1, sketched the architecture in Figure 2-1, and showed the input-mode routing in Figure 2-2. Supplementary rule identifiers are listed in Appendix A; synthetic broken inputs used to verify the validator's rule coverage are described in Appendix B.

3.1 Findings

Most UF formatting rules are enforced by the `ufdissertation` class, which means the validator's primary job is to detect overrides of the template's correct defaults rather than to re-implement the formatting rules themselves. A small set of rules genuinely require inspection of the rendered PDF; these are concentrated in the geometric and accessibility categories.

3.2 Limitations

The validator targets the Fall 2025+ UF template only. Older templates (accepted through Summer 2026) are detected and the validator exits with a clear message rather than producing potentially misleading output. Master's theses are out of scope for the initial release.

3.3 Future Work

Future releases will expand input surfaces: a Chrome extension for students writing in Overleaf, and a Visual Studio Code extension for students using a git-plus-editor workflow. External URL liveness checks, deferred from the initial release to keep the validator network-free, will be added behind an opt-in flag.

3.4 A Final Word

To the student who used this demo as a reference for their own dissertation: congratulations. The fact that you reached this section of this document means you are within reach of submission. Take a breath, work through the checklist in Appendix C, and submit when ready. The graduate school has seen better dissertations than yours and worse, and the difference between them was never the margins.

APPENDIX A SUPPLEMENTARY VALIDATOR RULES

This appendix lists rule identifiers and citations that did not warrant inline discussion in the main chapters but are checked by the validator. See `docs/uf-rules.md` for full detection strategies and source citations.

- UF-D1 – editMode option present
- UF-D2 – LuaLaTeX directive
- UF-D3 – overrideTitles / overrideChapters options
- UF-P1 – `\set*File` companions exist on disk

APPENDIX B SAMPLE BROKEN INPUTS

To verify that each must-fix rule fires, the test suite includes synthetic broken inputs that violate exactly one rule. Each lives in `tests/fixtures/broken/<rule-id>/` and includes a brief note explaining what was broken and why the corresponding rule should fire.

A clean release of the validator detects every broken fixture and flags it with the corresponding rule identifier.

APPENDIX C SUBMISSION CHECKLIST FOR THE STUDENT

The validator catches mechanical formatting violations. It does not catch everything. Before clicking submit, work through this checklist. Your committee, your department, and the UF Editorial Office will appreciate every box you mark.

C.1 Things the validator checks for you

- Margins, fonts, line spacing, alignment, paragraph indent (rules UF-F1 through UF-F7).
- Page numbering scheme (UF-F6).
- Required sections present and in order (UF-F8, UF-F9).
- At least three chapters (UF-F10).
- Required metadata macros set (UF-F14).
- Abstract under the 350-word cap (UF-F15).
- Broken internal references and uncited bibliography entries (UF-S3).
- Use of LuaLaTeX directive (UF-D2).

C.2 Things only you can check

- Your committee has signed the ETD signature page in GIMS.
- Your degree application is current in ONE.UF.
- Your dissertation title in the title page matches the title on your degree application.
- Your major matches your department's official major name.
- Your committee chair's name is spelled correctly. (And the co-chair's, if applicable.)
- The graduation month is May, August, or December and matches your intended graduation term.
- Every figure has a caption. Every table has a caption. Every equation that you cite has a label.
- Every citation in your text resolves to a real entry in your bibliography, and every bibliography entry is cited at least once.
- Your acknowledgments thank the people who actually got you here. (They will read it.)
- Co-authored chapters acknowledge co-authors.
- Published-chapter footnotes appear on the first page of each published-chapter, with the journal citation.

- Image resolution is high enough to read on screen.
- Hyperlinks in your PDF actually go where you expect.
- You have a backup of the source.
- You have a backup of the backup.
- You have eaten today.
- You have slept in the last twenty-four hours, ideally more than four.

C.3 One last note

You wrote a dissertation. Whatever happens at submission, that fact does not go away. Take a breath, run the checklist, and submit.

You earned this.

LIST OF REFERENCES

[1] University of Florida Graduate School Editorial Office. Formatting requirements for theses and dissertations, 2026. URL https://success.grad.ufl.edu/td/formatting/. Accessed 2026-05-27.

[2] University of Florida Information Technology Help Desk. Thesis and dissertation templates: MS Word and LaTeX, 2026. URL https://it.ufl.edu/helpdesk/graduate-resources/ms-word--latex-templates/. Template updated 2026-04-17; LuaLaTeX with TeX Live 2025 required.

BIOGRAPHICAL SKETCH

Jordan Avery Lee discovered LaTeX in their first semester of graduate school and discovered the UF formatting specification at 11:43 p.m. the night before their First Submission deadline. Both experiences shaped their dissertation.

Before graduate school, Lee earned a Bachelor of Science in Computer Science from a public research university whose graduate office, charitably, used Microsoft Word templates. Lee entered the doctoral program at the University of Florida the same year and devoted the next several years to research, teaching, two failed thesis title revisions, one successful committee reshuffle, and the cultivation of strong opinions about page breaks. They served as a graduate teaching assistant for three semesters and as a graduate research assistant for the remainder of their training, accruing along the way an unreasonable familiarity with the `titlesec` package.

Their research interests include static analysis, software tooling for academic workflows, and accessibility in technical documents. Their non-research interests include long walks, longer compile times, and the persistent suspicion that one more pass of `lualatex` will resolve the cross-reference warnings. (It usually does.)

Lee received the doctoral degree in May 2026. The dissertation compiled on the first try.[1]

[1]This is a lie; it compiled on the fourth try. The first three produced increasingly creative undefined references, but the dissertation is, after all, finished.